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CS 230 : Discrete Computational Structures

Summer Semester, 2025

HOMEWORK ASSIGNMENT #2

Due Date: Friday, June 6

Proofs

Suggested Reading: Lecture Notes Proofs 1 - 3; Rosen Sections 1.7 - 1.8; Lehman et al. Chapter 1

For the problems below, explain your answers and show your reasoning. To prove any properties of rational numbers, use the definition of a rational. Similarly, to prove properties of odd and even integers, use their definitions. You may not use the theorems that were proven in class, eg. the product of two odd integers is odd.

1. **[6 Pts]** Prove that for all integers p , p is even if and only if $p^3 + 3$ is odd. *Note: You need to do two proofs, one in each direction. You may not use the theorems shown in class regarding odd and even products.*

1) If p is even, then $p^3 + 3$ is odd

Assume: p is even

By definition of even: $p = 2k$

Substitute: $p^3 + 3 = (2k)^3 + 3 = 8k^3 + 3$

$8k^3$ is even (divisible by 2) and 3 is odd (not divisible by 2)

even + odd = odd

$p^3 + 3$ is odd

Therefore, if p is even, then $p^3 + 3$ is odd

2) If $p^3 + 3$ is odd, then p is even

Prove the contrapositive: If p is odd, then $p^3 + 3$ is even

Assume: p is odd

By definition of odd: $p = 2k + 1$

Substitute: $p^3 + 3 = (2k + 1)^3 + 3 = 8k^3 + 12k^2 + 6k + 4$

$8k^3 + 12k^2 + 6k + 4$ is even (divisible by 2)

Contrapositive Proven: If p is odd, then $p^3 + 3$ is even

Therefore, If $p^3 + 3$ is odd, then p is even

Since both side have be proven, for all p integers. p is even if and only if $p^3 + 3$ is odd.

2. [5 Pts] Prove that if x is irrational then $3x$ is irrational. What proof technique did you use and why?

Proof by Contrapositive

Claim: If x is irrational then $3x$ is irrational

Contrapositive: If $3x$ is rational then x is rational

Assume: $3x$ is rational and there exists integers y and $z \neq 0$ such that $3x = \frac{y}{z}$

$$x = \frac{y}{3z}$$

$$y \in \mathbb{Z}$$

$$3z \in \mathbb{Z}, z \neq 0$$

Therefore, x is a ratio of two integers and thus rational

Therefore, if $3x$ is rational then x is rational

Since the contrapositive was proven, if x is irrational, then $3x$ is irrational

I used proof by contrapositive because irrational numbers are difficult to prove. By taking the contrapositive of the given proof, I am given rational variables and it becomes much simpler to prove.

3. [6 Pts] Let m and n be positive integers. Prove that if $mn > 63$, then $m \geq 8$ or $n \geq 10$. What proof technique did you use and why?

Proof by Contrapositive

Claim: Let m and n be positive integers. If $mn > 63$, then $m \geq 8$ or $n \geq 10$.

Contrapositive: If $m < 8$ and $n < 10$, then $mn \leq 63$.

Since $m < 8$ and $n < 10$, the largest possible values for m and n are 7 and 9, respectively.

Thus, the maximum value of mn is: $mn \leq 7 \times 9 = 63$

Therefore, if $m < 8$ and $n < 10$, then $mn \leq 63$, which proves the contrapositive.

By the contrapositive, if $mn > 63$, then $m \geq 8$ or $n \geq 10$.

I used proof by contrapositive because it had a disjunction (or) and by using the contrapositive, that "or" will turn into an "and", making it easier to prove.

4. [6 Pts] Let x and y be rational numbers and let z be an irrational number. Prove that $xy + z$ is irrational. Can you use a direct proof? Why or why not?

Given:

$$x, y \in \mathbb{Q}$$

$$z \notin \mathbb{Q}$$

Proof:

Since x and y are rational, then product xy is rational.

The sum of a rational number and an irrational number is irrational.

Therefore, since xy is rational and z is irrational, the sum $xy + z$ must also be irrational.

Thus, $xy + z$ is irrational.

Direct Proof can be used as I just used it above to prove that $xy + z$ is irrational.

5. **[6 Pts]** Suppose your student organization has 26 scheduled meetings for the calendar year. Prove that at least three of the meetings fall on the same month. What proof technique do you use?

Claim: If there are 26 meetings scheduled throughout the year, then at least three meetings fall in the same month.

Proof by Contradiction:

Assume: No more than 2 meetings scheduled in any one month; 12 months in a year

Maximum number of meetings: $12 \times 2 = 24$ meetings

Given 26 meeting scheduled, thus: $26 > 24$

Therefore, the assumption must be false. Through proof by contradiction, there must be at least one month in which at least 3 meetings are scheduled.

6. [6 Pts] Prove that if $p \geq 5$ or $p \leq -3$ then $(p - 1)^2 \geq 16$. What proof technique did you use and why?

Claim: If $p \geq 5$ or $p \leq -3$, then $(p - 1)^2 \geq 16$.

Direct Proof:

Assume: $p \geq 5$ or $p \leq -3$.

For this assumption, there are two cases

Case 1: $p \geq 5$

Then $p - 1 \geq 4$

Squaring both sides

$$(p - 1)^2 \geq 4^2 = 16$$

Case 2: $p \leq -3$

Then $p - 1 \leq -4$

Squaring both sides

$$(p - 1)^2 \geq (-4)^2 = 16$$

In both cases, $(p - 1)^2 \geq 16$. Therefore, if $p \geq 5$ or $p \leq -3$, then $(p - 1)^2 \geq 16$.

I used direct proof because it was the most simplest and logical method of proving the given proposition.

7. [5 Pts] Prove that there exist irrational numbers x and y whose sum is rational. Is your proof constructive or non-constructive? Explain.

Claim: There exist irrational numbers x and y whose sum is rational.

Proof:

Let:

$$x = \sqrt{2}$$

$$y = -\sqrt{2}$$

Both x and y are irrational numbers since the negative of an irrational number is irrational.

$$x + y = \sqrt{2} + (-\sqrt{2}) = 0$$

Since 0 is a rational number, I have found two irrational numbers whose sum is rational.

This is a constructive proof, because I explicitly provided two irrational numbers x and y , and showed that their sum is rational.

Sets

Suggested Reading: Rosen Sections 2.1 - 2.3; Lehman et al. Chapter 4.1, 4.3, 4.4.

For the problems below, explain your answers and show your reasoning.

8. [6 Pts] Let A and B be non-empty sets. Prove that if $A \neq B$, then $A \times B \neq B \times A$.

Proof by Contrapositive: If $A \times B = B \times A$, then $A = B$.

Assume: $A \times B = B \times A$.

For all $a \in A$ and $b \in B$, the ordered pair (a, b) is in $B \times A$.

But since $(a, b) \in B \times A$ implies that $a \in B$ and $b \in A$,

Every $a \in A$ must also be in $B \Rightarrow A \subseteq B$

Every $b \in B$ must also be in $A \Rightarrow B \subseteq A$

Hence, $A = B$

Since the contrapositive is true, then if $A \neq B$, then $A \times B \neq B \times A$ is also true

9. [6 Pts] Prove that $(A \cup B) - C = (A - C) \cup (B - C)$ using iff arguments and logical equivalences.

Prove: $(A \cup B) - C = (A - C) \cup (B - C)$ by showing that both sides are subsets of each other

Case 1: $(A \cup B) - C \subseteq (A - C) \cup (B - C)$

Let $x \in (A \cup B) - C$.

Definition of set difference: $x \in A \cup B$ and $x \notin C$

Definition of union: $x \in A$ or $x \in B$

From this I got $x \in A$ and $x \notin C$, so $x \in A - C$ or $x \in B$ and $x \notin C$, so $x \in B - C$

Therefore, $x \in (A - C) \cup (B - C)$

Thus, $(A \cup B) - C \subseteq (A - C) \cup (B - C)$.

Case 2: $(A - C) \cup (B - C) \subseteq (A \cup B) - C$

Let $x \in (A - C) \cup (B - C)$.

Definition of union: $x \in A - C$ or $x \in B - C$

I get either $x \in A$ and $x \notin C$ or $x \in B$ and $x \notin C$

In both cases: $x \in A \cup B$ and $x \notin C$

Therefore, $x \in (A \cup B) - C$

Thus, $(A - C) \cup (B - C) \subseteq (A \cup B) - C$.

Since both subsets are equal: $(A \cup B) - C = (A - C) \cup (B - C)$

10. [10 Pts] Disprove the statements below.

(a) If $A \cup C \subseteq B \cup C$ then $A \subseteq B$.

Counterexample:

Let: $A = \{1\}$, $B = \{2\}$, $C = \{1, 2\}$

$A \cup C = \{1\} \cup \{1, 2\} = \{1, 2\}$

$B \cup C = \{2\} \cup \{1, 2\} = \{1, 2\}$

Therefore, $A \cup C \subseteq B \cup C$ is true

However, $A = \{1\}$, $B = \{2\} \Rightarrow A \not\subseteq B$

Therefore, the statement is false since A is not a subset of B .

(b) If $A \cap C \subseteq B \cap C$ then $A \subseteq B$.

Counterexample:

Let: $A = \{1, 2\}$, $B = \{2\}$, $C = \{2\}$

$A \cap C = \{1, 2\} \cap \{2\} = \{2\}$

$B \cap C = \{2\} \cap \{2\} = \{2\}$

Therefore, $A \cap C \subseteq B \cap C$ is true

However, $A = \{1, 2\}$, $B = \{2\} \Rightarrow A \not\subseteq B$

Therefore, the statement is false since A is not a subset of B .

11. [8 Pts] Prove by contradiction that $(A - C) \cap (C - B) = \emptyset$.

Proof by Contradiction:

Assume: $(A - C) \cap (C - B) \neq \emptyset$

There exists an element x such that: $x \in (A - C) \cap (C - B)$

Definition of set difference and intersection:

$x \in A$ and $x \notin C$ (since $x \in A - C$)

$x \in C$ and $x \notin B$ (since $x \in C - B$)

But then $x \in C$ and $x \notin C$, which is a contradiction.

Therefore, the assumption must be false, and therefore, $(A - C) \cap (C - B) = \emptyset$

12. [8 Pts] Prove that if $A \cup C \subseteq B \cup C$ and $A \cap C \subseteq B \cap C$ then $A \subseteq B$. *Hint: You can either do a proof by contradiction or a proof by cases.*

Proof by Cases:

Assume: $x \in A$.

I want to show that $x \in B$ so cases will proceed on whether $x \in C$ or not.

Case 1: $x \in C$

$x \in A \cap C$, since $x \in A$ and $x \in C$.

Given that $A \cap C \subseteq B \cap C$, it follows that $x \in B \cap C$, so $x \in B$.

Case 2: $x \notin C$

$x \in A$, and since $x \notin C$, I still have $x \in A \cup C$.

Given that $A \cup C \subseteq B \cup C$, it follows that $x \in B \cup C$.

But $x \notin C$, so $x \in B$.

In either case, $x \in B$. Since $x \in A$ was arbitrary, $A \subseteq B$ end up being proven true.

13. [12 Pts] Prove that $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$ using subset argument. You *may not* use logical equivalences in your proof. Use general proof techniques like ‘proof by contradiction’ and ‘proof by cases’.

Proof by Cases:

(1) I have to show that $(A \cup B) - (A \cap B) \subseteq (A - B) \cup (B - A)$

Assume: $x \in (A \cup B) - (A \cap B)$.

Then $x \in A \cup B$ and $x \notin A \cap B$.

So x is in at least one of A or B , but not both.

Case 1: $x \in A$ and $x \notin B$

Then $x \in A - B$, so $x \in (A - B) \cup (B - A)$

Case 2: $x \in B$ and $x \notin A$

Then $x \in B - A$, so $x \in (A - B) \cup (B - A)$

In both cases, $x \in (A - B) \cup (B - A)$.

Therefore, $(A \cup B) - (A \cap B) \subseteq (A - B) \cup (B - A)$

(2) I have to show that $(A - B) \cup (B - A) \subseteq (A \cup B) - (A \cap B)$

Assume: $x \in (A - B) \cup (B - A)$.

Case 1: $x \in A - B$

Then $x \in A$ and $x \notin B$.

So $x \in A \cup B$ and $x \notin A \cap B$.

Thus, $x \in (A \cup B) - (A \cap B)$

Case 2: $x \in B - A$

Then $x \in B$ and $x \notin A$.

So $x \in A \cup B$ and $x \notin A \cap B$.

Thus, $x \in (A \cup B) - (A \cap B)$

Therefore, $(A - B) \cup (B - A) \subseteq (A \cup B) - (A \cap B)$

Since both inclusions hold, $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$